

REG3A Polymorphism Is Associated with the Incidence of Extensive Chronic Gvhd after Allogeneic BMT

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Abstract

[Introduction] Regenerating islet-derived protein 3 alpha (REG3A), a protein produced by Paneth cells, has been reported as a biomarker of lower gastrointestinal graft-versus-host disease (GVHD). The single nucleotide polymorphism (SNP) rs7588571 is located at the upstream of REG3A gene within 2kb, suggesting that the expression of REG3A might be affected by the SNP. In this study, we analyzed the association between the rs7588571 genotype in recipients and transplant outcomes.

[Patients and methods] The study population included 118 adult Japanese patients who had undergone T cell replete allogeneic bone marrow transplantation from an HLA-identical sibling donor with GVHD prophylaxis consisting of short-term methotrexate and cyclosporine between 1987 and 2006. Sixty-eight (58%) male and 50 (42%) female patients, with a median age of 35 years (range 15-56), were included. Eighty-eight (75%) patients were classified as having standard risk disease defined as acute myeloid leukemia in first or second complete remission, acute lymphoblastic leukemia without Philadelphia chromosome in first complete remission, chronic myeloid leukemia in chronic phase, myelodysplastic syndrome without excess of blasts, aplastic anemia and paroxysmal nocturnal hemoglobinuria, while other patients were classified as having high risk disease. Conditioning regimen was myeloablative in 113 (96%) patients and reduced-intensity in other patients. Genomic DNA was extracted from patient's

peripheral blood or bone marrow by standard methods. Genotyping was performed using TaqMan SNP Genotyping Assays (Applied Biosystems, Foster City, California, USA) on ABI 7300 Real-Time PCR systems (Applied Biosystems). Multiple logistic regression model was used to identify risk factors associated with acute and chronic GVHD, and multivariate Cox model was created for overall survival (OS), non-relapse mortality (NRM), and relapse. Variables with a P-value < 0.1 in the univariate analysis were entered into the stepwise selection method. All tests were two-sided, and P < 0.05 was considered significant. This study was approved by the ethics committees at the Nagoya University Hospital and the Japanese Red Cross Nagoya Daiichi Hospital.

[Results] The frequencies of the GG, AG, and AA genotypes were 41%, 43%, and 16%, respectively, which were compatible with those reported in the HapMap-JPT database. There was no significant association between the distribution pattern of the rs7588571 genotypes and each patient characteristic. Grade II-IV acute GVHD developed in 12% of all patients, 14% in those with GG genotype, 14% with AG genotype and 11% with AA genotype, respectively. In univariate analysis, the higher patient age (≥ 35) was the only significant factor for higher incidence of grade II-IV acute GVHD (OR: 5.4, 95% CI: 1.3-32, P=0.011). Extensive chronic GVHD developed in 38% of all patients, 27% in those with GG genotype, 45% with AG genotype and 50% with AA genotype, respectively. The incidence of extensive chronic GVHD in non-GG genotype (46%) was significantly higher than that in GG genotype (27%) (P=0.05). In multivariate analysis, non-GG genotype (OR: 2.5, 95% CI: 1.0-5.8, P=0.04), female to male transplant (2.8, 1.2-7.0, 0.023), and high risk disease (2.8, 1.1-7.2, 0.038) were significant factors for higher incidence of extensive chronic GVHD. The relapse rates at 4-year were 26% in non-GG and 17% in GG genotype (HR: 1.16, 95% CI: 0.49-2.7, P=0.73). High risk disease was the only significant factor for higher relapse rate. The OS and NRM at 4-year were 57% and 20% in non-GG, and 65% and 17% in GG genotype, respectively. The rs7588571 genotypes affected neither OS nor NRM.

[Discussion] REG3A polymorphism was associated with the incidence of extensive chronic GVHD. Given its capacity to control the tissue regeneration and intestinal microbiota, REG3A may not only reflect the intestinal damage caused by GVHD but also have a certain role in the pathogenesis of GVHD. Our findings need to be confirmed in larger cohort, and further investigation is required in order to clarify the biological effect of rs7588571 genotypes and the exact role of REG3A in the pathophysiology of GVHD.

Disclosures

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Author notes

*Asterisk with author names denotes non-ASH members.

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